

BANARAS HINDU UNIVERSITY

B.A. (Hons.) — Semester IV Examination, 2023-24

Paper No. MTB-401: Partial Differential Equations

Time: 3 Hours

Full Marks: 70

Ref. No.: 076465

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Derive the PDE by eliminating arbitrary constants from the equation of a sphere.
- (b) Form a PDE by eliminating ϕ from $z = (x - y)\phi(x^2 + y^2)$.
- (c) Solve: $p \tan x + q \tan y = \tan z$.
- (d) Find the complete integral of $pq = p + q$.
- (e) Find the particular integral of $r + s - 2t = e^{x+y}$.
- (f) Solve: $(D - D')z = 0$.
- (g) Solve: $(2D - 3)(3D - 5D' - 7)^2 z = 0$.

Question 2.

- (a) Find the PDE by eliminating a, b, c from $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. [7]
- (b) Find the complete integral, general integral, and singular integral of $z = yp^2 - 4$. [7]

Question 3.

- (a) Find the integral surface of $x^2p + y^2q + z^2 = 0$ passing through the hyperbola $xy = x + y$, $z = 1$. [7]
- (b) Solve: $(D^2 - DD' - 2D'^2)z = (y - 1)e^x$. [7]

Question 4.

- (a) Solve: $(D^2 + DD' + D' - 1)z = \sin(x + 2y)$. [7]
- (b) Solve: $ys + p = \cos(x + y) - y \sin(x + y)$. [7]

Question 5.

(a) Solve: $(x^2D^2 + 2xyDD' + y^2D'^2)z = x^m y^n$ where $(m + n) \neq 0, 1$. [7]

(b) Solve: $y^2(x - y)p + x^2(y - x)q = z^2(x^2 + y^2)$. [7]

Question 6.

(a) Reduce $r - t = 0$ to canonical form. [7]

(b) Solve $r = a^2t$ by Monge's method. [7]

Question 7.

(a) Describe the two-dimensional heat equation by separation of variables. [7]

(b) Solve the wave equation by separation of variables:

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = x^2(25 - x^2)$$

[7]